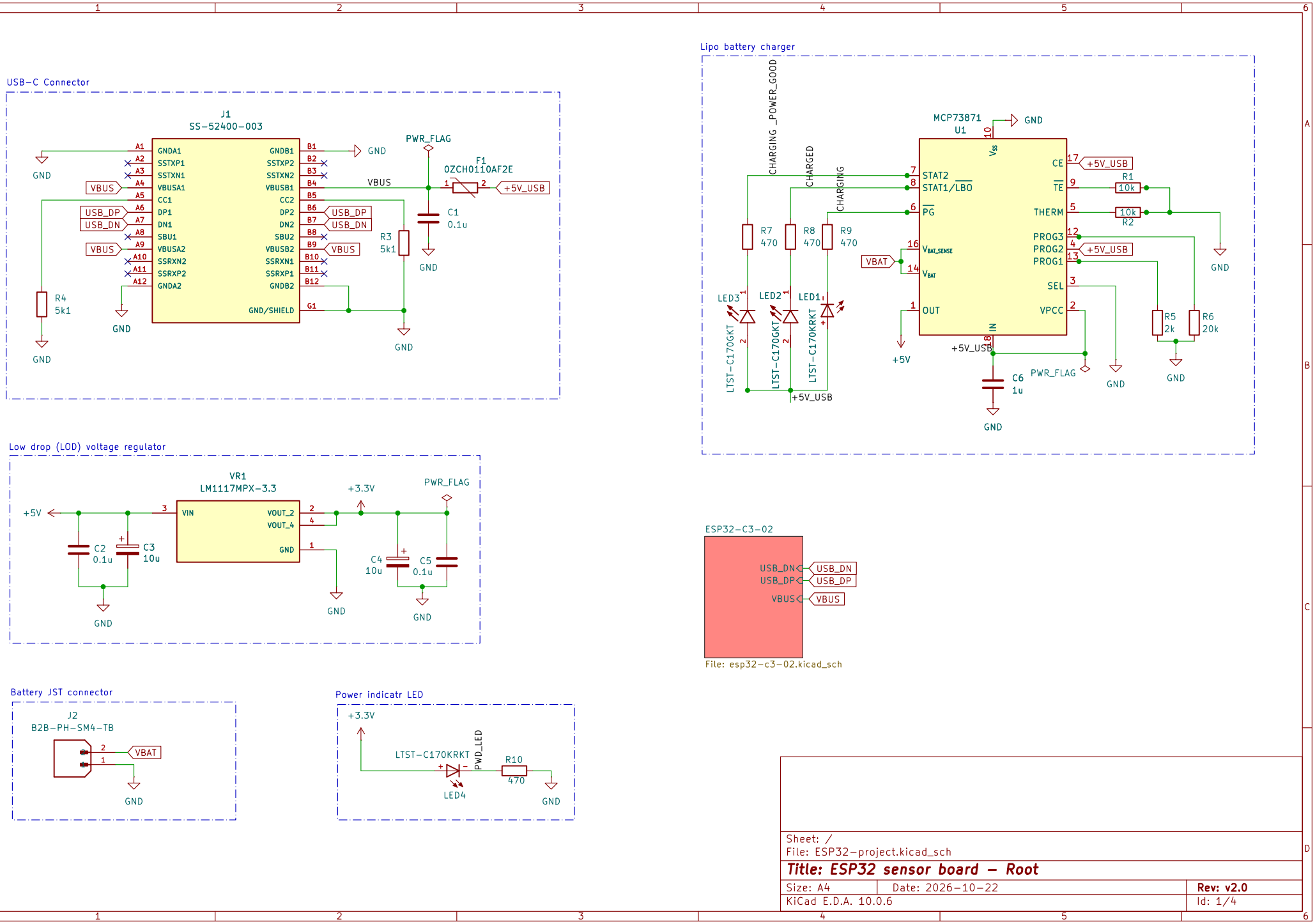
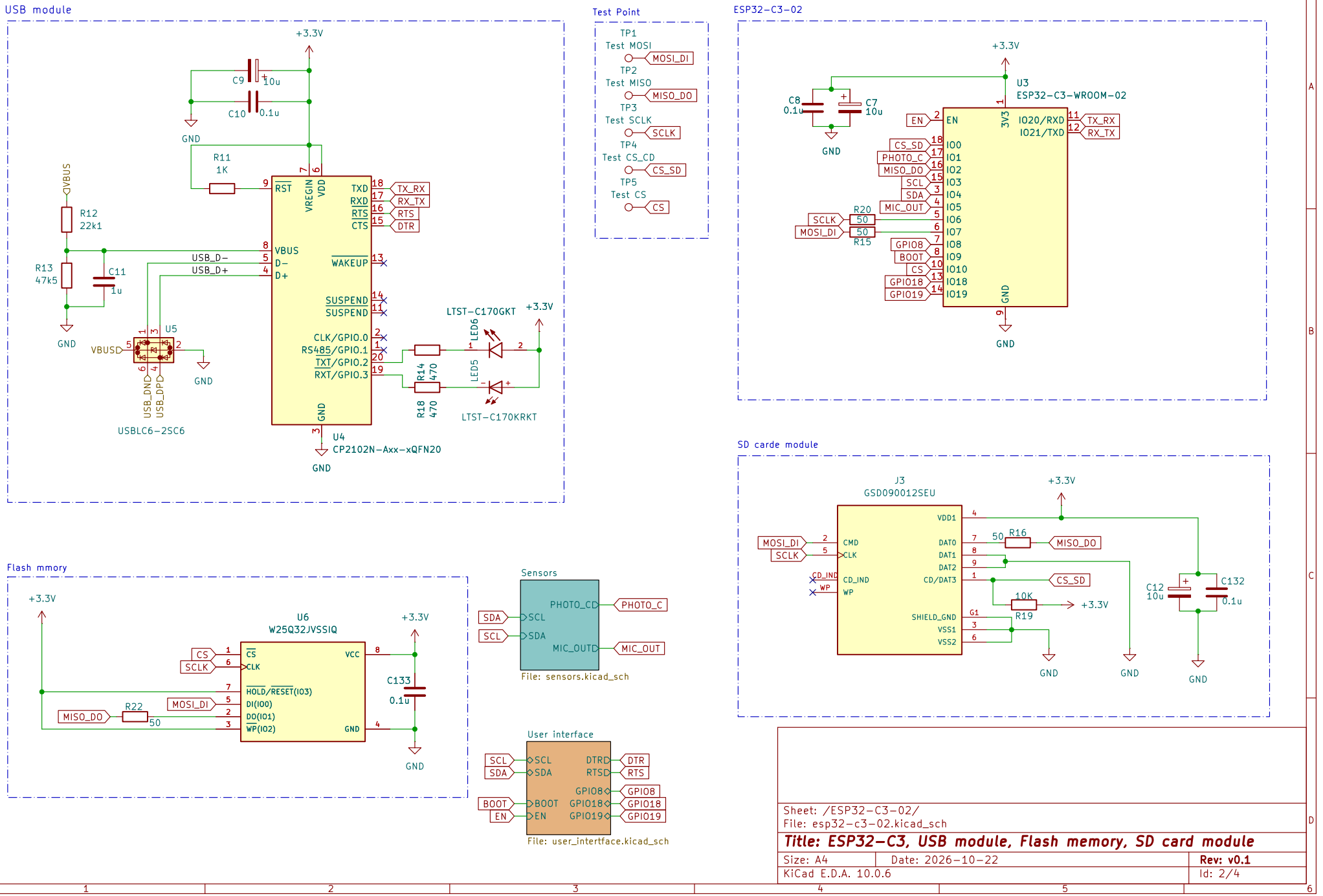
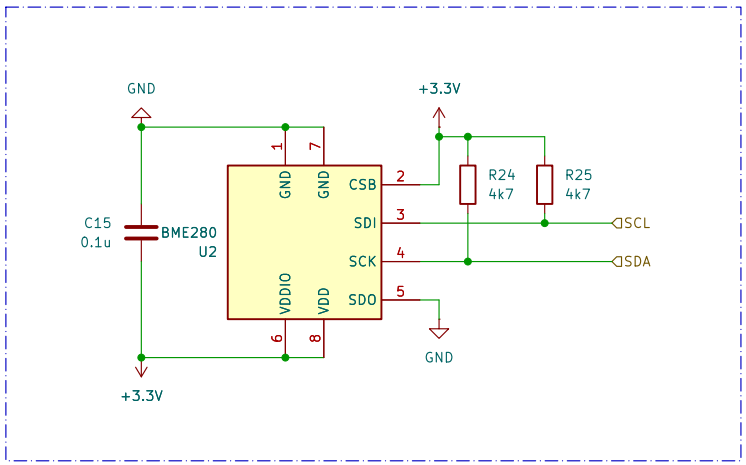

USB-C Connector
J1
SS-52400-003
GNDA1
SSTXP1
SSTXN1
VBUSA1
CC1
DP1
DN1
SBU1
VBUSA2
SSRXN2
SSRXP2
GNDA2
GNDB1
SSTXP2
SSTXN2
VBUSB1
CC2
DP2
DN2
SBU2
VBUSB2
SSRXN1
SSRXP1
GNDB2
GND/SHIELD
VBUS
USB_DP
USB_DN
GND
PWR_FLAG
F1
0ZCH0110AF2E
+5V_USB
C1
0.1u
R3
5k1
R4
5k1
Lipo battery charger
MCP73871
U1
CHARGING_POWER_GOOD
CHARGED
CHARGING
STAT2
STAT1/LBO
PG
VBAT_SENSE
VBAT
OUT
Vss
VIN
CE
TE
THERM
PROG3
PROG2
PROG1
SEL
VPCC
R1
10k
R2
10k
R5
2k
R6
20k
R7
470
R8
470
R9
470
LED1
LED2
LED3
LTST-C170KRKT
LTST-C170GKT
+5V
C6
1u
Low drop (LOD) voltage regulator
VR1
LM1117MPX-3.3
VIN
VOUT_2
VOUT_4
GND
+5V
+3.3V
C2
0.1u
C3
10u
C4
10u
C5
0.1u
ESP32-C3-02
USB_DN
USB_DP
VBUS
File: esp32-c3-02.kicad_sch
Battery JST connector
J2
B2B-PH-SM4-TB
VBAT
Power indicatr LED
LTST-C170KRKT
PWD_LED
R10
470
LED4
Sheet: /
File: ESP32-project.kicad_sch
Title: ESP32 sensor board - Root
Size: A4
Date: 2026-10-22
Rev: v2.0
KiCad E.D.A. 10.0.6
Id: 1/4


USB module
+3.3V
C9
10u
C10
0.1u
GND
R11
1K
VBUS
R12
22k1
R13
47k5
C11
1u
USB_D-
USB_D+
RST
VREGIN
VDD
VBUS
D-
D+
WAKEUP
SUSPEND
SUSPEND
CLK/GPIO.0
RS485/GPIO.1
TXT/GPIO.2
RXT/GPIO.3
GND
TXD
RXD
RTS
CTS
TX_RX
RX_TX
RTS
DTR
U5
VBUSD
USB_DND
USB_DPD
USBLC6-2SC6
U4
CP2102N-Axx-xQFN20
LTST-C170GKT
LED6
LED5
R14
470
R18
470
LTST-C170KRKT
Test Point
TP1
Test MOSI
MOSI_DI
TP2
Test MISO
MISO_DO
TP3
Test SCLK
SCLK
TP4
Test CS_CD
CS_SD
TP5
Test CS
CS
ESP32-C3-02
U3
ESP32-C3-WROOM-02
C8
0.1u
C7
10u
EN
3V3
IO20/RXD
IO21/TXD
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
R20
50
R15
50
SD carde module
J3
GSD090012SEU
CMD
CLK
CD_IND
WP
VDD1
DAT0
DAT1
DAT2
CD/DAT3
SHIELD_GND
VSS1
VSS2
R16
50
10K
R19
C12
10u
C132
0.1u
Flash mmory
U6
W25Q32JVSSIQ
CS
SCLK
MISO_DO
MOSI_DI
R22
50
HOLD/RESET(IO3)
DI(IO0)
DO(IO1)
WP(IO2)
VCC
C133
0.1u
Sensors
SDA
SCL
PHOTO_CD
MIC_OUT
PHOTO_C
File: sensors.kicad_sch
User interface
DTRD
RTSD
BOOT
EN
File: user_intertface.kicad_sch
Sheet: /ESP32-C3-02/
File: esp32-c3-02.kicad_sch
Title: ESP32-C3, USB module, Flash memory, SD card module
Size: A4
Date: 2026-10-22
Rev: v0.1
KiCad E.D.A. 10.0.6
Id: 2/4

BME280



Microphone and Amplifer

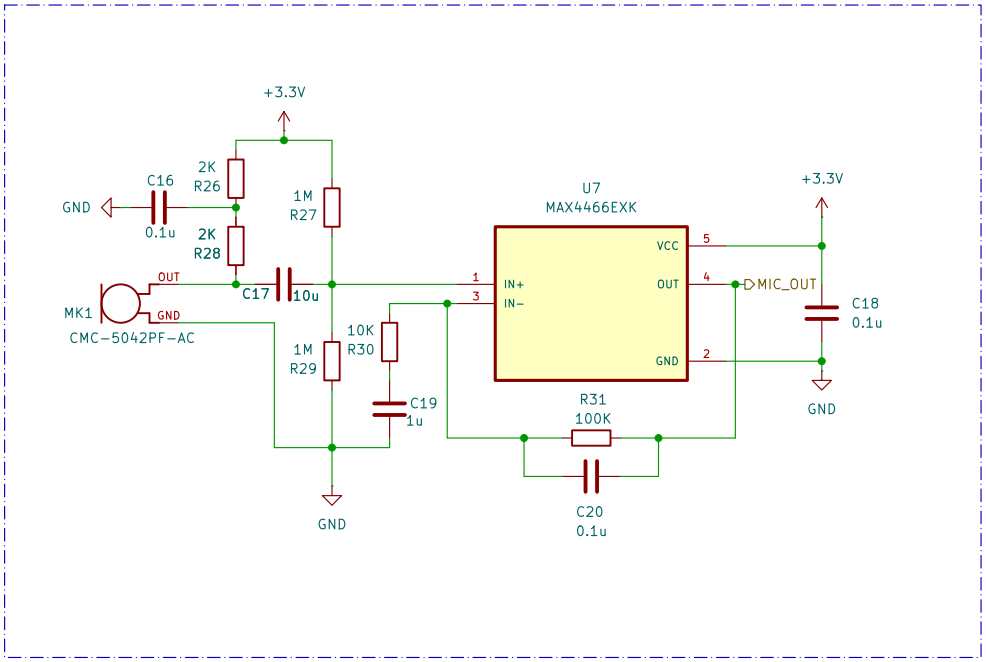
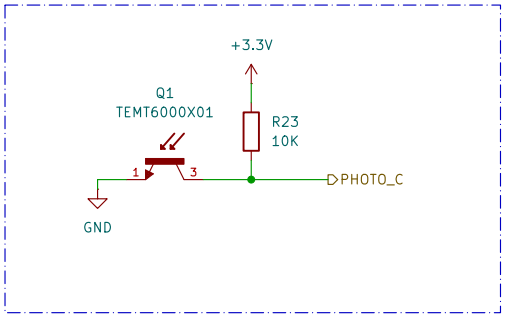
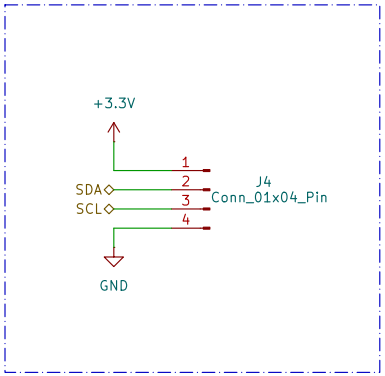


Photo sensor

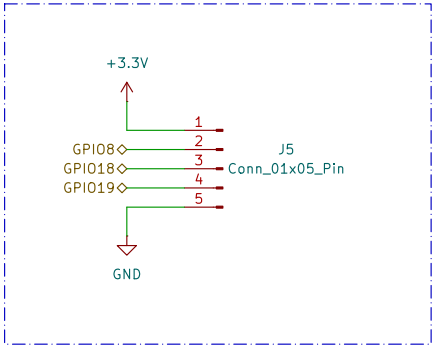


Sheet: /ESP32-C3-02/Sensors/		
File: sensors.kicad_sch		
Title: BME280, Photo sensor, Microphone and Amplifer		
Size: A4	Date: 2026-10-23	**Rev: v1.0**
KiCad E.D.A. 10.0.6		Id: 3/4

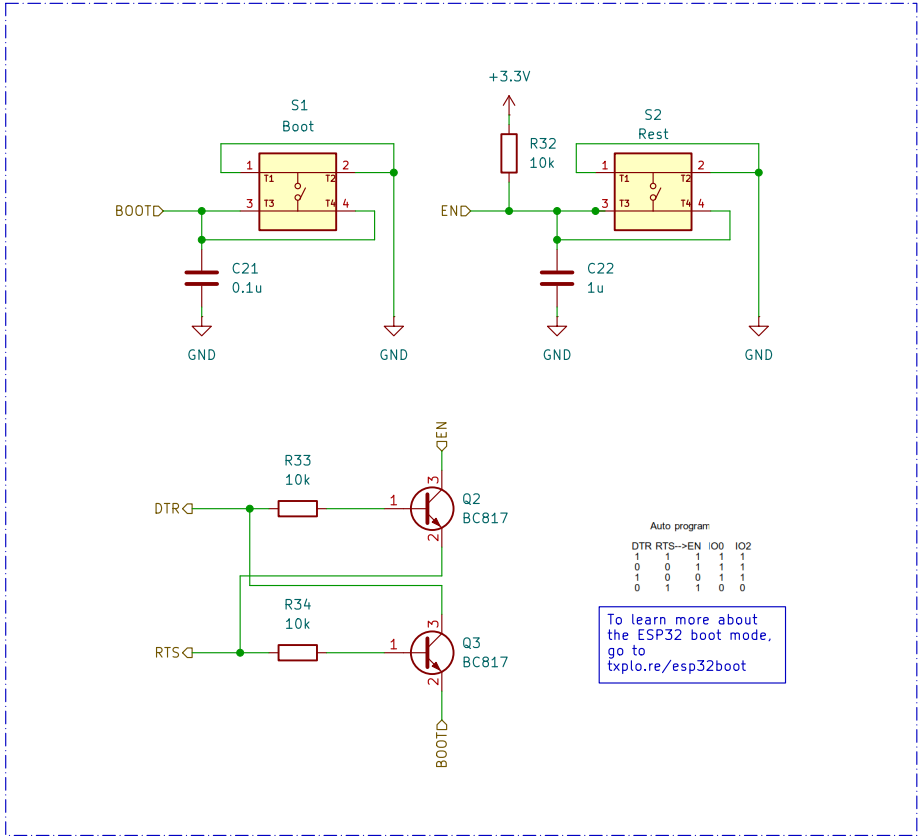
OLED display connector (I2C)



Breakout for remaining pins



REST and BOOT buttons



Sheet: /ESP32-C3-02/User interface/		
File: user_intertface.kicad_sch		
Title: OLED display (I2C) pin,REST and BOOT buttons,Breakout pins		
Size: A4	Date: 2026-10-23	**Rev: v1.0**
KiCad E.D.A. 10.0.6		Id: 4/4